(ii) Move the decimal point by the number of places specified by the power of 10, to the right if positive, or to the left if negative. Add zeros if necessary.

(iii) Rewrite the number in decimal form.

Example 2.29 Write the following numbers in decimal form:

(i) 6.34×10^4 (ii) 2.00367×10^{-5}

Solution

(i) 6.34×10^4

$\Rightarrow \quad 6 \quad . \quad 3 \quad 4 \quad 0 \quad 0 \quad = 63400$



(ii) 2.00367×10^{-5}

$0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 2 \quad . 0 0 3 6 7$



$= 0.0000200367$

2.8.3 Arithmetic of Numbers in Scientific Notation

(i) If the indices in the scientific notation of two numbers are the same, addition (or subtraction) is easily performed.

Example 2.30 The mass of the Earth is 5.97×10^{24} kg and that of the Moon is 0.073×10^{24} kg. What is their total mass?

Solution

$$\begin{aligned} \text{Total mass} &= 5.97 \times 10^{24} \text{ kg} + 0.073 \times 10^{24} \text{ kg} \\ &= (5.97 + 0.073) \times 10^{24} \text{ kg} \\ &= 6.043 \times 10^{24} \text{ kg} \end{aligned}$$

(ii) The product or quotient of numbers in scientific notation can be easily done if we make use of the laws of radicals appropriately.

Example 2.31 Write the following in scientific notation :

(i) $\left(50000000\right)^4$ (ii) $\left(0.00000005\right)^3$

(iii) $\left(300000\right)^3 \times \left(2000\right)^4$ (iv) $\left(4000000\right)^3 \div \left(0.00002\right)^4$

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